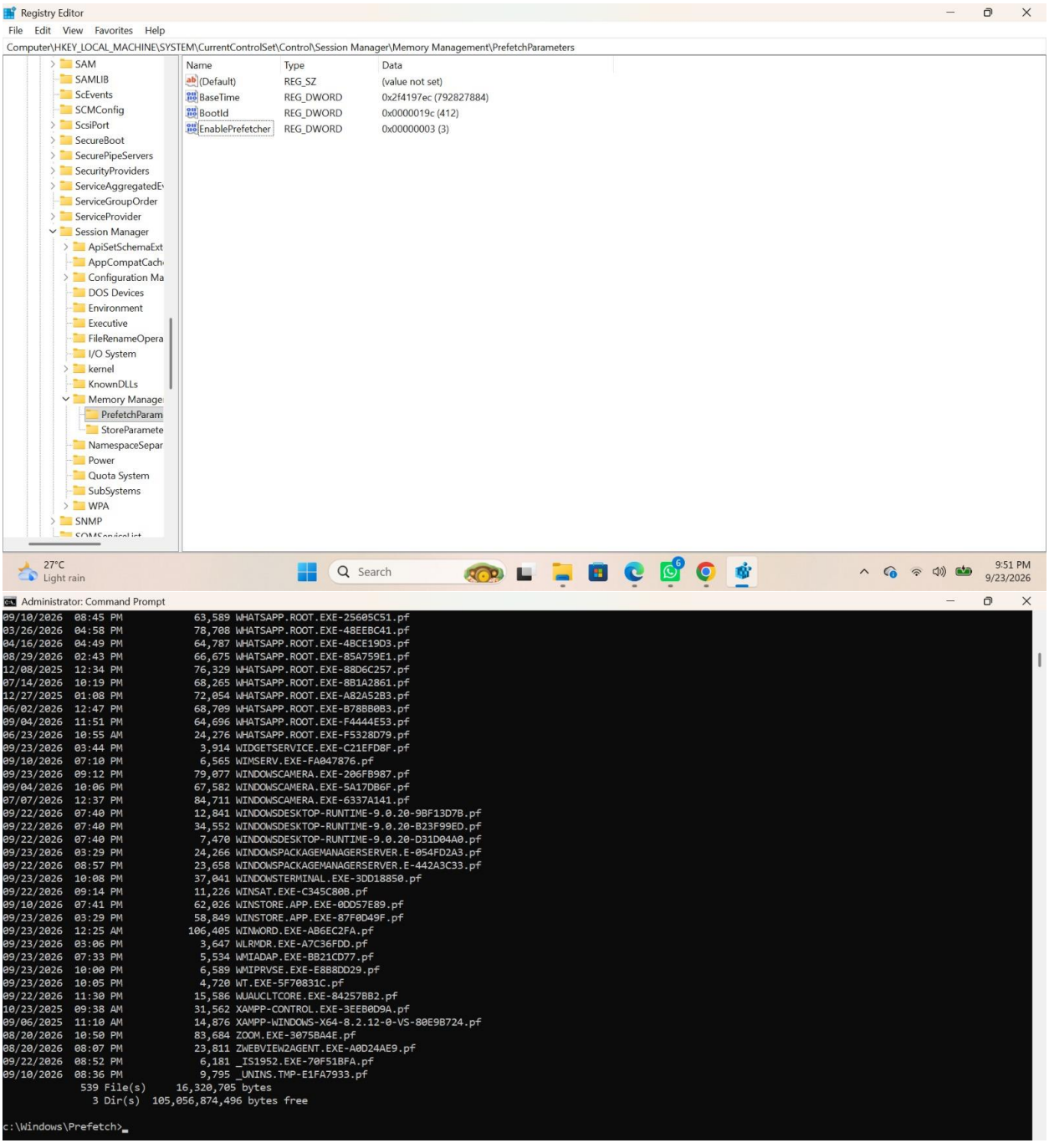
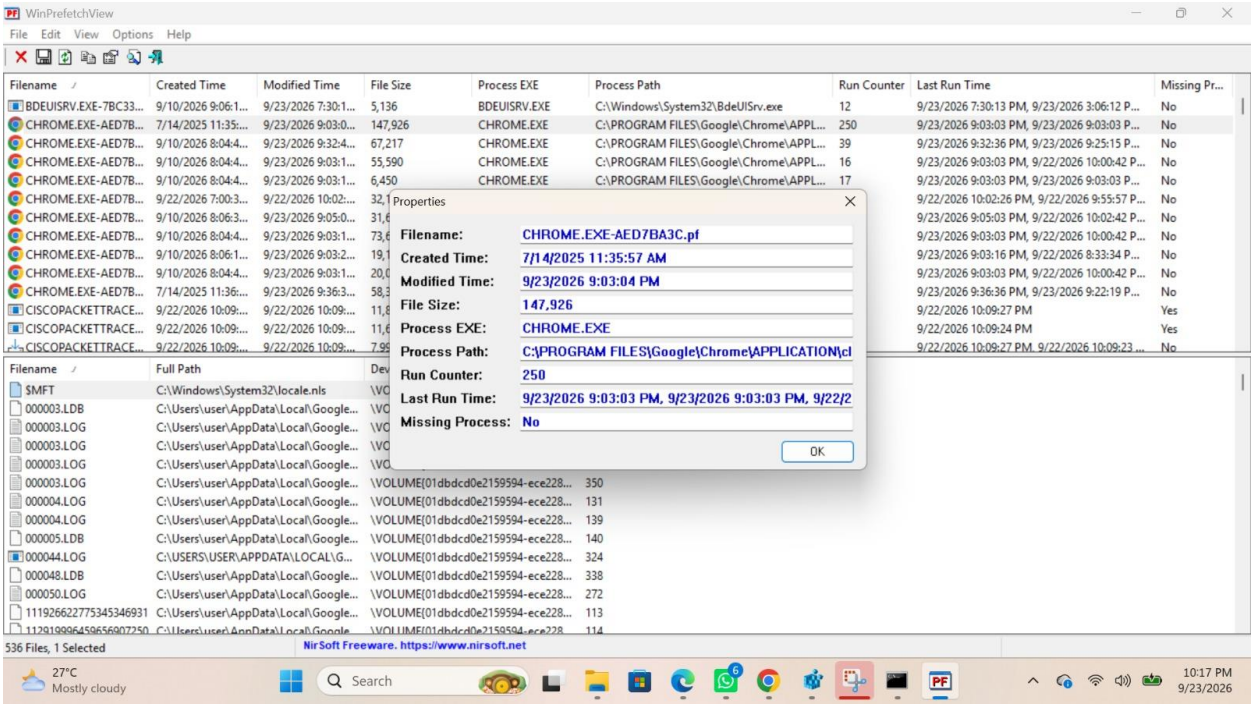
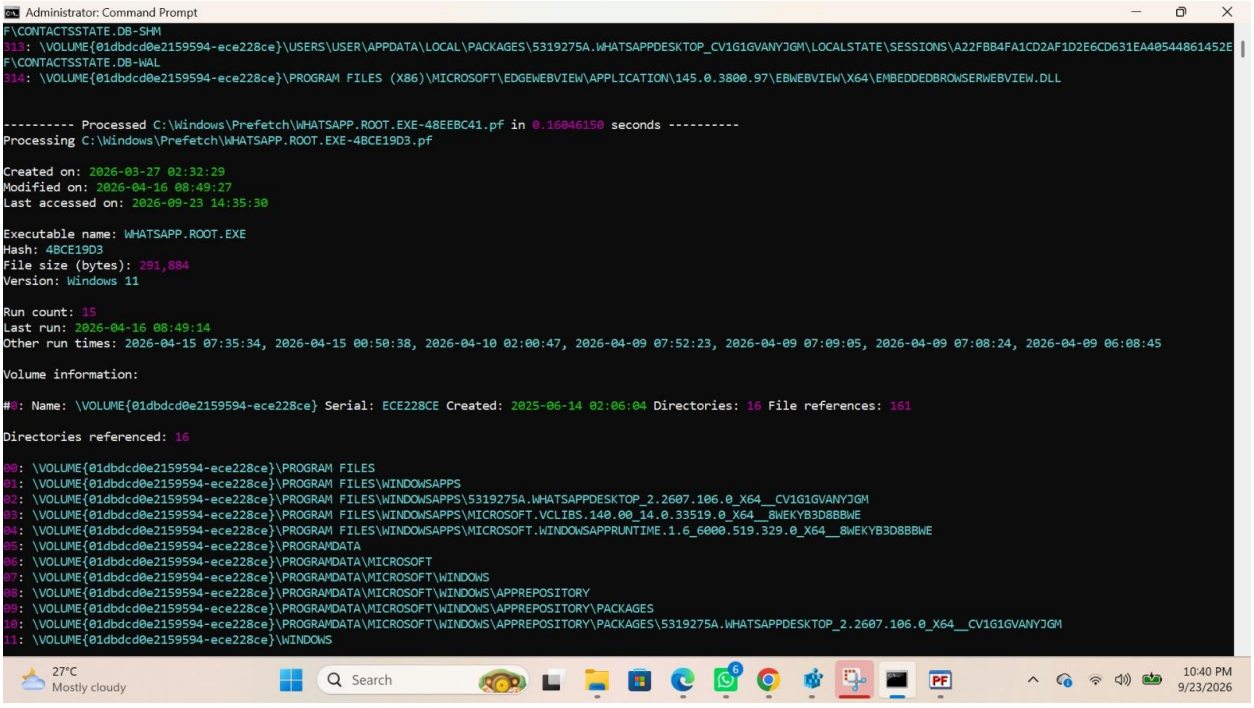
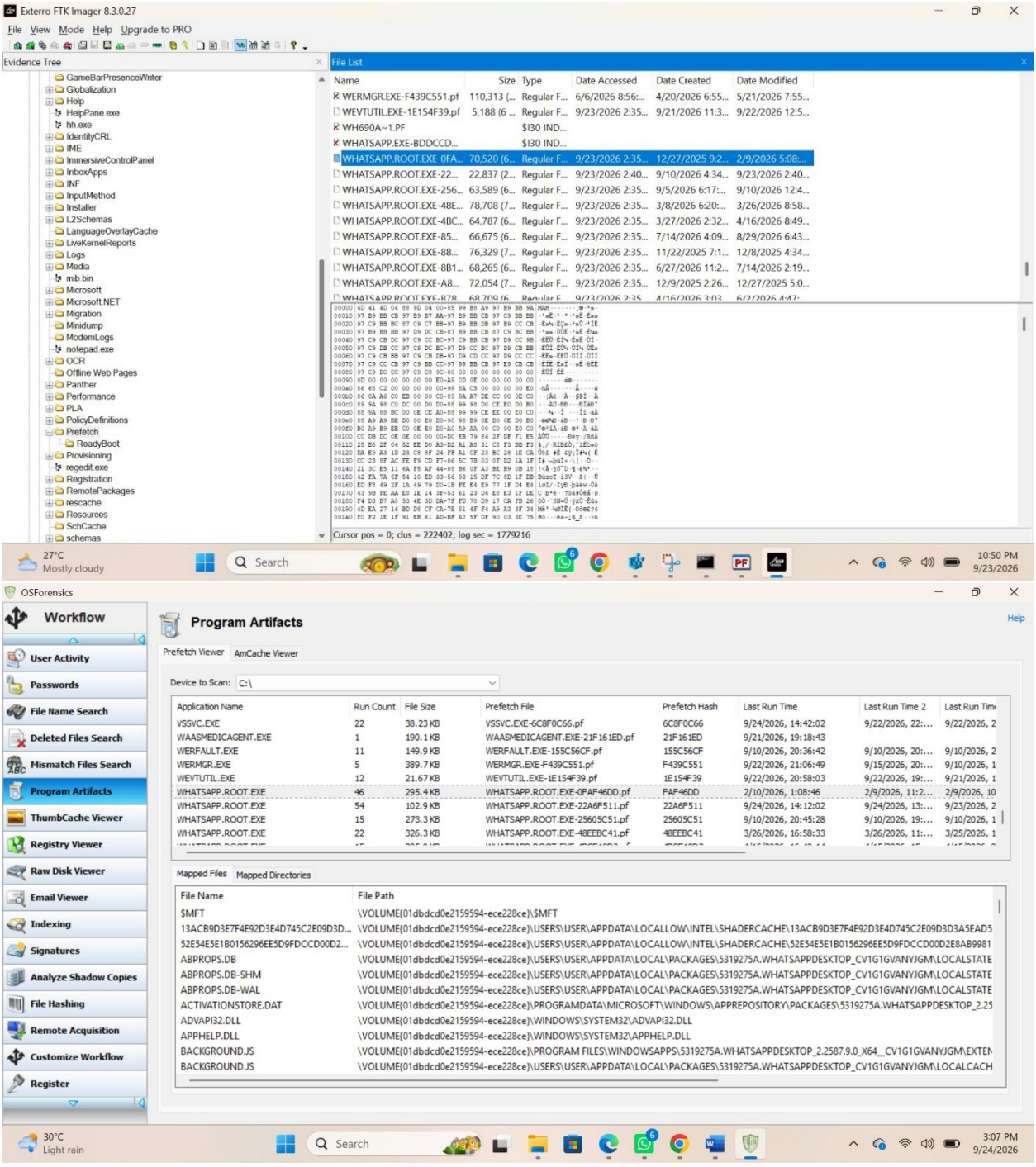
2 Lab Report: Forensic Investigation of Windows Prefetch Files




WinPrefetchView
File Edit View Options Help
Filename | Created Time | Modified Time | File Size | Process EXE | Process Path | Run Counter | Last Run Time | Missing Pr...
BDEUISRV.EXE-7BC33... | 9/10/2026 9:06:1... | 9/23/2026 7:30:1... | 5,136 | BDEUISRV.EXE | C:\Windows\System32\BdeUISrv.exe | 12 | 9/23/2026 7:30:13 PM, 9/23/2026 3:06:12 P... | No
CHROME.EXE-AED7B... | 7/14/2025 11:35:... | 9/23/2026 9:03:0... | 147,926 | CHROME.EXE | C:\PROGRAM FILES\Google\Chrome\APPL... | 250 | 9/23/2026 9:03:03 PM, 9/23/2026 9:03:03 P... | No
CHROME.EXE-AED7B... | 9/10/2026 8:04:4... | 9/23/2026 9:32:4... | 67,217 | CHROME.EXE | C:\PROGRAM FILES\Google\Chrome\APPL... | 39 | 9/23/2026 9:32:36 PM, 9/23/2026 9:25:15 P... | No
CHROME.EXE-AED7B... | 9/10/2026 8:04:4... | 9/23/2026 9:03:1... | 55,590 | CHROME.EXE | C:\PROGRAM FILES\Google\Chrome\APPL... | 16 | 9/23/2026 9:03:03 PM, 9/22/2026 10:00:42 P... | No
CHROME.EXE-AED7B... | 9/10/2026 8:04:4... | 9/23/2026 9:03:1... | 6,450 | CHROME.EXE | C:\PROGRAM FILES\Google\Chrome\APPL... | 17 | 9/23/2026 9:03:03 PM, 9/23/2026 9:03:03 P... | No
CHROME.EXE-AED7B... | 9/22/2026 7:00:3... | 9/22/2026 10:02:... | 32,1 | | | | 9/22/2026 10:02:26 PM, 9/22/2026 9:55:57 P... | No
CHROME.EXE-AED7B... | 9/10/2026 8:06:3... | 9/23/2026 9:05:0... | 31,6 | | | | 9/23/2026 9:05:03 PM, 9/22/2026 10:02:42 P... | No
CHROME.EXE-AED7B... | 9/10/2026 8:04:4... | 9/23/2026 9:03:1... | 73,6 | | | | 9/23/2026 9:03:03 PM, 9/22/2026 10:00:42 P... | No
CHROME.EXE-AED7B... | 9/10/2026 8:06:1... | 9/23/2026 9:03:2... | 19,1 | | | | 9/23/2026 9:03:16 PM, 9/22/2026 8:33:34 P... | No
CHROME.EXE-AED7B... | 9/10/2026 8:04:4... | 9/23/2026 9:03:1... | 20,0 | | | | 9/23/2026 9:03:03 PM, 9/22/2026 10:00:42 P... | No
CHROME.EXE-AED7B... | 7/14/2025 11:36:... | 9/23/2026 9:36:3... | 58,3 | | | | 9/23/2026 9:36:36 PM, 9/23/2026 9:22:19 P... | No
CISCOPACKETTRACE... | 9/22/2026 10:09:... | 9/22/2026 10:09:... | 11,8 | | | | 9/22/2026 10:09:27 PM | Yes
CISCOPACKETTRACE... | 9/22/2026 10:09:... | 9/22/2026 10:09:... | 11,6 | | | | 9/22/2026 10:09:24 PM | Yes
CISCOPACKETTRACE... | 9/22/2026 10:09:... | 9/22/2026 10:09:... | 7,95 | | | | 9/22/2026 10:09:27 PM, 9/22/2026 10:09:23 ... | No
Properties
Filename: CHROME.EXE-AED7BA3C.pf
Created Time: 7/14/2025 11:35:57 AM
Modified Time: 9/23/2026 9:03:04 PM
File Size: 147,926
Process EXE: CHROME.EXE
Process Path: C:\PROGRAM FILES\Google\Chrome\APPLICATION\cl
Run Counter: 250
Last Run Time: 9/23/2026 9:03:03 PM, 9/23/2026 9:03:03 PM, 9/22/2
Missing Process: No
OK
Filename | Full Path | Dev
$MFT | C:\Windows\System32\locale.nls | \VO
000003.LDB | C:\Users\user\AppData\Local\Google... | \VO
000003.LOG | C:\Users\user\AppData\Local\Google... | \VO
000003.LOG | C:\Users\user\AppData\Local\Google... | \VO
000003.LOG | C:\Users\user\AppData\Local\Google... | \VO
000003.LOG | C:\Users\user\AppData\Local\Google... | \VOLUME{01dbdcd0e2159594-ece228... | 350
000004.LOG | C:\Users\user\AppData\Local\Google... | \VOLUME{01dbdcd0e2159594-ece228... | 131
000004.LOG | C:\Users\user\AppData\Local\Google... | \VOLUME{01dbdcd0e2159594-ece228... | 139
000005.LDB | C:\Users\user\AppData\Local\Google... | \VOLUME{01dbdcd0e2159594-ece228... | 140
000044.LOG | C:\USERS\USER\APPDATA\LOCAL\G... | \VOLUME{01dbdcd0e2159594-ece228... | 324
000048.LDB | C:\Users\user\AppData\Local\Google... | \VOLUME{01dbdcd0e2159594-ece228... | 338
000050.LOG | C:\Users\user\AppData\Local\Google... | \VOLUME{01dbdcd0e2159594-ece228... | 272
11192662275345346931 | C:\Users\user\AppData\Local\Google... | \VOLUME{01dbdcd0e2159594-ece228... | 113
536 Files, 1 Selected
NirSoft Freeware. https://www.nirsoft.net
27°C Mostly cloudy
Search
10:17 PM 9/23/2026


Administrator: Command Prompt
F\CONTACTSSTATE.DB-SHM
313: \VOLUME{01dbdcd0e2159594-ece228ce}\USERS\USER\APPDATA\LOCAL\PACKAGES\5319275A.WHATSAPPDESKTOP_CV1G1GVANYJGM\LOCALSTATE\SESSIONS\A22FBB4FA1CD2AF1D2E6CD631EA40544861452E
F\CONTACTSSTATE.DB-WAL
314: \VOLUME{01dbdcd0e2159594-ece228ce}\PROGRAM FILES (X86)\MICROSOFT\EDGEWEBVIEW\APPLICATION\145.0.3800.97\EBWEBVIEW\X64\EMBEDDEDBROWSERWEBVIEW.DLL
---------- Processed C:\Windows\Prefetch\WHATSAPP.ROOT.EXE-48EEBC41.pf in 0.16046150 seconds ----------
Processing C:\Windows\Prefetch\WHATSAPP.ROOT.EXE-4BCE19D3.pf
Created on: 2026-03-27 02:32:29
Modified on: 2026-04-16 08:49:27
Last accessed on: 2026-09-23 14:35:30
Executable name: WHATSAPP.ROOT.EXE
Hash: 4BCE19D3
File size (bytes): 291,884
Version: Windows 11
Run count: 15
Last run: 2026-04-16 08:49:14
Other run times: 2026-04-15 07:35:34, 2026-04-15 00:50:38, 2026-04-10 02:00:47, 2026-04-09 07:52:23, 2026-04-09 07:09:05, 2026-04-09 07:08:24, 2026-04-09 06:08:45
Volume information:
#0: Name: \VOLUME{01dbdcd0e2159594-ece228ce} Serial: ECE228CE Created: 2025-06-14 02:06:04 Directories: 16 File references: 161
Directories referenced: 16
00: \VOLUME{01dbdcd0e2159594-ece228ce}\PROGRAM FILES
01: \VOLUME{01dbdcd0e2159594-ece228ce}\PROGRAM FILES\WINDOWSAPPS
02: \VOLUME{01dbdcd0e2159594-ece228ce}\PROGRAM FILES\WINDOWSAPPS\5319275A.WHATSAPPDESKTOP_2.2607.106.0_X64__CV1G1GVANYJGM
03: \VOLUME{01dbdcd0e2159594-ece228ce}\PROGRAM FILES\WINDOWSAPPS\MICROSOFT.VCLIBS.140.00_14.0.33519.0_X64__8WEKYB3D8BBWE
04: \VOLUME{01dbdcd0e2159594-ece228ce}\PROGRAM FILES\WINDOWSAPPS\MICROSOFT.WINDOWSAPPRUNTIME.1.6_6000.519.329.0_X64__8WEKYB3D8BBWE
05: \VOLUME{01dbdcd0e2159594-ece228ce}\PROGRAMDATA
06: \VOLUME{01dbdcd0e2159594-ece228ce}\PROGRAMDATA\MICROSOFT
07: \VOLUME{01dbdcd0e2159594-ece228ce}\PROGRAMDATA\MICROSOFT\WINDOWS
08: \VOLUME{01dbdcd0e2159594-ece228ce}\PROGRAMDATA\MICROSOFT\WINDOWS\APPREPOSITORY
09: \VOLUME{01dbdcd0e2159594-ece228ce}\PROGRAMDATA\MICROSOFT\WINDOWS\APPREPOSITORY\PACKAGES
10: \VOLUME{01dbdcd0e2159594-ece228ce}\PROGRAMDATA\MICROSOFT\WINDOWS\APPREPOSITORY\PACKAGES\5319275A.WHATSAPPDESKTOP_2.2607.106.0_X64__CV1G1GVANYJGM
11: \VOLUME{01dbdcd0e2159594-ece228ce}\WINDOWS
27°C Mostly cloudy
Search
10:40 PM 9/23/2026



1. Introduction

The Windows Prefetch files (.pf) are features that have been created in order to speed up the time it takes to boot the system and to decrease the latency associated with starting applications. Each time an application is run, Windows keeps a record of this event by storing important forensic metadata, including the number of times it has been executed, the timestamps, and the resources that have been loaded. In the field of digital forensics, these prefetch files act as definitive proof

of program execution, even if the original executable file has been uninstalled or deleted by the suspect.

2. Prefetch Configuration and System Baseline

It's necessary to check the prefetch status in the Windows Registry before carrying out the analysis to make sure that execution logging is enabled.

- **Registry Key:** `Computer\HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Session Manager\Memory Management\PrefetchParameters`
- **Target Value:** `EnablePrefetcher`
- **Observed Value:** `0x00000003 (3)`
- **What is certain is that a value of 3 indicates** both **Application Prefetching** (with a value of 1) and **Boot Prefetching** (with a value of 2) being fully active.

3. Identification of Artifacts of the Native File System

By default, prefetch records are stored in `%SystemRoot%\Prefetch` (`C:\Windows\Prefetch`). The directory's contents were examined using an elevated Command Prompt.

- **Path:** `C:\Windows\Prefetch`
- **Naming Structure:** `<EXECUTABLE_NAME>-<HASH>.pf`
- **Summary Count:** 539 files verified on disk
- The 8-character hash is derived from the absolute path of the executable, thus allowing it to tell apart binaries that have the same name but are executed from different folders.

4. The analysis of forensic artifacts using a multi-tool approach

4.1 NirSoft WinPrefetchView

The WinPrefetchView program was used to read and prioritize the metadata of the `.pf` files via a graphical interface without changing the files' timestamps.

- **Target Artifact:** `CHROME.EXE-AED7BA3C.pf`
- **Process Path:** `C:\PROGRAM FILES\Google\Chrome\APPLICATION\chrome.exe`
- **Run Count:** 250 executions

- **Last Run Time:** `9/23/2026 9:03:03 PM`
- **Missing Process:** `No` (confirms the binary still exists at the location)

4.2 Eric Zimmerman's PECmd (Prefetch Explorer Command Line)

The Windows prefetch file structure was parsed and decompressed using PECmd in order to obtain the execution details.

- **Command Run: run** `PECmd.exe with the parameter -d and the path "C:\Windows\Prefetch"`
- **Target Artifact:** `WHATSAPP.ROOT.EXE-4BCE19D3.pf`
- **Path Hash:** `4BCE19D3`
- **Run Count:** 15
- **Last Run Time:** `2026-04-16 08:49:14`
- **The historical run times include 7 more previous** execution timestamps (making a total of 8).
- **There are 16 directories that have been referenced**.

4.3 PassMark OSForensics

The Program Artifacts module of OSForensics was used in order to carry out a cross-examination of the prefetch records and to check the application dependencies.

- **Module:** Program Artifacts (Prefetch Viewer)
- **The target artifacts are** `WHATSAPP.ROOT.EXE` (`0FAF46DD` and `22A6F511`).
- **Observed Run Counts:** 46 and 54 executions
- **The system dependencies have been mapped, the mapped files comprising** system binaries (`ADVAPI32.DLL`, `APPHELP.DLL`), SQLite databases (`ABPROPS.DB`), and application scripts (`BACKGROUND.JS`).

4.4 AccessData FTK Imager

The FTK Imager was used to mount the logical drive (`C:\`) in order to examine the raw file system allocation and hex structure of the prefetch folder.

- **Directory Viewed:** `C:\Windows\Prefetch`
- **The raw hex header at o**ffset `0000` is `4D 41 4D 04` (the ASCII signature is **`MAM`**).
- **The fact that the MAM signature is present serves to confirm the Windows LZNT1 compression format and ensures that the file's fundamental structure is verified** directly from the disk sectors.

5. A summary table of the findings extracted

Forensic Feature	WinPrefetchView	PECmd	OSForensics	FTK Imager
Analyzed Binary	`CHROME.EXE`	`WHATSAPP.ROOT.EXE`	`WHATSAPP.ROOT.EXE`	`WHATSAPP.ROOT.EXE`
Path Hash	`AED7BA3C`	`4BCE19D3`	`0FAF46DD/22A6F511`	In directory index

|
| **Run Count** | 250

| 15

| 46 / 54

| Present in data stream

|
| **Primary Timestamp** | `9/23/2026 9:03:03 PM`|`2026-04-16 08:49:14`|`9/24/2026 14:12:02`|`2/9/2026 5:08 PM`|
| **Key Extraction** | Verified binary path

| 8 historical run times

| Loaded DLL/file paths

| Raw `MAM` hex signature

|

---

6. Conclusion

The Windows Prefetch records offer reliable, multi-layered evidence of the way programmes were executed. When the evidence obtained from various GUI triage tools (such as WinPrefetchView), automated parsers (like PECmd), digital forensic packages (such as OSForensics), and direct file system viewers (FTK Imager) is cross-checked, investigators are able to reconstruct the timeline of application executions, verify the total number of times the programmes were launched, and identify the software dependencies that were loaded.